

The Accreditation Problem: Continuous-Validity Authorization for Agentic Systems (v1.1 errata)

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v1.1-preprint

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Abstract

The accreditation model that classified-information systems have inherited from NIST SP 800-37 and adjacent doctrine is a **snapshot** model: a system is accredited at a point in time, on the basis of evidence gathered up to that point, and the accreditation remains valid until the next scheduled review.¹ For agentic systems on the tactical substrate (EDO-P1), the snapshot model is structurally inadequate. The system evolves continuously: fine-tunes apply, personas update, envelopes adjust, substrate state drifts. Accreditation must therefore become **continuous-validity authorization**: an architectural commitment that the accreditation evidence is generated, maintained, and verified continuously, and that the accreditation’s validity at any moment is a runtime computable property.^{2,3} This paper specifies the archi-

¹National Institute of Standards and Technology, *Risk Management Framework for Information Systems and Organizations*, NIST SP 800-37 Rev. 2 (Gaithersburg, MD, 2018), <https://doi.org/10.6028/NIST.SP.800-37r2>.

²Nicolas Chaillan and Mark Tatum, “Continuous Authority to Operate (cATO): The DoD’s Path to Software Agility,” *IEEE Software* 39, no. 1 (2022): 12-18, <https://doi.org/10.1109/MS.2021.3127155>.

³National Institute of Standards and Technology, *Information Security Continuous Monitoring (ISCM) for Federal Information Systems and Organizations*, NIST SP 800-137 (Gaithersburg, MD, 2011),

ture. Existing draft material in drafts/edo-wave-2/P9-accreditation-problem.md informs the present draft.

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1. Introduction

NIST SP 800-37 established the Risk Management Framework (RMF) as the dominant accreditation methodology for U.S. federal and DoD systems.⁴ The framework's value is well-established; its application to agentic systems requires translation, particularly under DDIL conditions where the snapshot model's underlying assumptions weaken. Continuous-monitoring extensions (NIST SP 800-137) and continuous-ATO research have moved the RMF toward runtime validity verification.⁵⁶ This paper applies that direction to agentic operation specifically.

§2 names five failure modes of snapshot-accreditation for agentic systems. §3 specifies continuous-validity architecture. §4 treats validity as continuous. §5 binds to Skipjack. §6 returns to HGC³AE².

2. Failure Modes of Snapshot Accreditation for Agentic Systems

Drift between accreditation and runtime. The accredited configuration diverges from the running configuration over time; the accreditation is operationally stale before the next scheduled review.

Fine-tune outside the boundary. A fine-tune applied between accreditations produces a model that has not been accredited; the runtime accreditation status is ambiguous.

Persona-update without re-accreditation. Persona prompt, scaffolding, or tool-manifest changes alter the system's behavior in ways the accreditation evidence did not address.

Envelope-expansion ambiguity. When the governance layer expands the envelope, is the accreditation still valid? The snapshot model has no clean answer.

Reconnection-evidence gap. When a substrate reconnects after partition, the accreditation evidence accumulated during partition may not have been transmitted; the receiving accreditation system reasons against a stale evidence set.

<https://doi.org/10.6028/NIST.SP.800-137>.

⁴National Institute of Standards and Technology, *Risk Management Framework for Information Systems and Organizations*, NIST SP 800-37 Rev. 2 (Gaithersburg, MD, 2018), <https://doi.org/10.6028/NIST.SP.800-37r2>.

⁵National Institute of Standards and Technology, *Information Security Continuous Monitoring (ISCM) for Federal Information Systems and Organizations*, NIST SP 800-137 (Gaithersburg, MD, 2011), <https://doi.org/10.6028/NIST.SP.800-137>.

⁶Nicolas Chaillan and Mark Tatum, "Continuous Authority to Operate (cATO): The DoD's Path to Software Agility," *IEEE Software* 39, no. 1 (2022): 12-18, <https://doi.org/10.1109/MS.2021.3127155>.

3. Continuous-Validity Authorization Architecture

Accredited-baseline as runtime property. The current accreditation baseline is a runtime-queryable property: every operationally significant component (model, persona, envelope, configuration) carries a reference to the baseline against which it is accredited.⁷

Continuous evidence pipeline. Evidence required for accreditation — configuration state, test results, audit logs, attestation chains — is generated continuously by the operational substrate and aggregated to the accreditation authority. The pipeline does not require periodic batched evidence gathering.⁸

Re-accreditation triggers. Explicit events trigger re-accreditation review: fine-tune deployment, persona update, envelope expansion, substrate reconfiguration. The triggers are enumerated; the agentic system surfaces each event as it occurs.

Authority chain. The accreditation authority (DAA / AO in DoD vocabulary) holds responsibility for each accreditation decision and for re-authorization at trigger events. The architecture surfaces decisions to the authority; it does not auto-accredit.

Compliance-as-code. Accreditation rules are encoded in machine-checkable form (compliance-as-code) so that the runtime can verify configuration against accreditation rules without batched human review.⁹

4. Continuous Validity as Process

Three phases. **Generation** of evidence continuously by the operational substrate. **Verification** continuously against the accreditation baseline. **Re-authorization** on trigger events, with the authority chain explicitly invoked.

5. Skipjack Coupling

Skipjack mechanisms operate over the continuous-validity surface.¹⁰ **Context integrity** is enforced through accreditation baseline verification. **Structured routing** uses accreditation status as a routing input. **HITL control points** fire on re-authorization triggers. **Iterative feedback** closes via accreditation telemetry feeding governance ceremony.

6. Implications and Conclusion

The implications extend to compliance posture (continuous-ATO is operationally feasible and architecturally specifiable), tooling (compliance-as-code and continuous-monitoring

⁷Nicolas Chaillan and Mark Tatum, “Continuous Authority to Operate (cATO): The DoD’s Path to Software Agility,” *IEEE Software* 39, no. 1 (2022): 12–18, <https://doi.org/10.1109/MS.2021.3127155>.

⁸Atul Adya et al., “Slicer: Auto-Sharding for Datacenter Applications,” *Proceedings of the 12th USENIX Symposium on Operating Systems Design and Implementation (OSDI ’16)*, 2016, <https://www.usenix.org/conference/osdi16/technical-sessions/presentation/adya>. (Architecturally relevant evidence-pipeline pattern; supplemented by compliance-as-code research below.)

⁹Niloofar Reza, Mauro Conti, and Saqib Hakak, “Compliance-as-Code: A Survey,” *ACM Computing Surveys* 56, no. 8 (2024): article 200, <https://doi.org/10.1145/3651308>.

¹⁰Justin H. Kuiper, CISSP, *Skipjack Protocol* (v1.0-preprint, 2026), zenodo.org/records/19869293.

pipelines), and authority structures (DAA/AO engagement at trigger events rather than periodic reviews).

The core claim: **accreditation for agentic systems is a continuous-validity property, not a snapshot; the architecture specifies how the property is computable at runtime and re-authorized at meaningful events.**

Returned to HGC³AE² letter by letter:

- **H — Human-governed.** Accreditation authority (DAA/AO) is human; the architecture surfaces decisions to that authority.
- **G — Curated Context.** Accreditation baseline and evidence pipeline are curated context.
- **C¹ — Cybersecurity.** Accreditation is the security-baseline anchor.
- **C² — Continuity.** Continuity of accreditation is the paper's commitment.
- **C³ — Coherence.** Accreditation baseline coherent with current operating configuration.
- **A — Agentially Engineered.** Evidence generation is agentic; authority decisions are human.
- **E — Executed.** Continuous verification at runtime.
- The exponent — the closure — is the generate-verify-re-authorize loop.

The remaining EDO papers (interoperability, deployment) operate on the accreditation surface this paper specifies.¹¹¹²

References

— end v1.0-preprint-draft —

Drafting state: v1.0-preprint draft. PR count: 12 inline (NIST SP 800-37 · NIST SP 800-137 · Chaillan IEEE Software · Adya OSDI · Reza ACM CSUR · Kelly DSN · Shahin IEEE Access · Khan C&S · Sandhu IEEE Computer · Havelund ASE · Bloomfield SCS + supporting). Existing draft drafts/edo-wave-2/P9-accreditation-problem.md folds into v0.5 expansion.

Appendix — v1.1 Errata Addendum

Added 2026-05-17 per Round-2 EDITORIAL-STAMP HOLD-substance verdict. The v1.0-preprint remains canonical at its original DOI; this v1.1 supplements with the corrections below.

§X. v1.1 Errata — Round-2 Substance Corrections

§X.1. Why this addendum exists

EDO-P9, *The Accreditation Problem: Continuous-Validity Authorization for Agentic Systems*, shipped as v1.0-preprint on 2026-05-01 at DOI 10.5281/zenodo.20180170, reader-visible on nonsequitur.tech/pubs/white-papers/accreditation-problem/. The paper received its Round-1 editorial stamp (1 of 3) on 2026-05-12.

¹¹EDO-P10 promotion (in concurrent drafting; yks-build#109).

¹²EDO-P11 CAPSTONE promotion (in concurrent drafting; yks-build#110).

On 2026-05-17, the Round-2 second-pass review (stamper: Edna; parent thread: YKS-Spine-Binder#355) returned a verdict of HOLD-substance. The review identified four substance defects and a citation/reference cluster issue in the published v1.0 text. Because v1.0 is in the wild with an active DOI, the remediation path is this errata addendum rather than a silent revision. v1.0 remains live and citable; v1.1 ships as a companion record naming what changed and why.

The defects, in summary: (1) §4 is substantively thin — three sentences restating §3 as process phases without operational animation; (2) §5’s Skipjack coupling is label-affixed rather than load-bearing — it declares mappings without specifying what continuous-validity changes about each Skipjack mechanism; (3) the [^evidence-pipeline] citation stretches a datacenter auto-sharding paper to anchor an accreditation evidence-pipeline claim without making the architectural analogy explicit; (4) seven references appear in the reference list but are never cited in the body text, inflating the apparent citation count. Each is corrected below.

§X.2. Defect: §4 process-phase thinness

v1.0 said: “Three phases. **Generation** of evidence continuously by the operational substrate. **Verification** continuously against the accreditation baseline. **Re-authorization** on trigger events, with the authority chain explicitly invoked.” (§4, complete text.)

The problem: §4 restates §3’s five architectural commitments as a three-phase process list without animating how the process executes on the substrate. The section is shape without substance. The Round-2 stamp finding (§4 substance audit) noted the contrast with EDO-P1’s treatment of substrate lifecycle, which walks through concrete operational sequences, and identified two acceptable remediation paths: (a) a concrete operational walkthrough showing what happens at a specific trigger event, or (b) an honest scope note deferring the process animation to implementation-level treatment.

Canonical correction: §4 adopts remediation path (a). The corrected text replaces the three-sentence list with a concrete operational walkthrough anchored to a single trigger event — fine-tune deployment — that traces the generate-verify-re-authorize loop end to end:

§4. Continuous Validity as Process

The architecture in §3 specifies five commitments. This section animates them as a single operational sequence, using fine-tune deployment as the exemplar trigger event.

Generation. The deployment pipeline emits a structured attestation record at fine-tune application: model identifier, training-data provenance hash, resulting weight checksum, timestamp, and the accreditation-baseline version against which the prior model was accredited. This record enters the continuous evidence pipeline (§3) without batched gathering — the pipeline ingests it at deployment time. The attestation schema is compliance-as-code: the fields required are specified in the accreditation-rule encoding, and the pipeline validates completeness before forwarding.¹³

Verification. The accreditation baseline (§3, runtime-queryable property) is read. The new model’s attestation record is evaluated against the baseline’s accreditation rules. Two outcomes are possible: (1) the fine-tune falls within the accredited envelope — e.g., the training data, hyperparameters, and resulting behavior profile match the categories the baseline already covers — in which case the verification step records a pass and the model enters service under the existing accreditation; or (2) the fine-tune exceeds the accredited envelope, in which case verification records

¹³Nilofar Reza, Mauro Conti, and Saqib Hakak, “Compliance-as-Code: A Survey,” *ACM Computing Surveys* 56, no. 8 (2024): article 200, <https://doi.org/10.1145/3651308>.

a boundary-exceeded finding and the model does NOT enter service. The model is held pending re-authorization.¹⁴

Re-authorization. On a boundary-exceeded finding, the authority chain (§3) is invoked. The accreditation authority (DAA/AO) receives the attestation record, the verification finding, and the specific envelope boundary that was exceeded. The authority renders a re-authorization decision: approve (expanding the baseline to cover the new envelope), reject (the fine-tune does not deploy), or conditionally approve (deploy with additional monitoring constraints). The decision is recorded as a new accreditation-baseline entry, and the baseline version increments. The HITL control point here is not optional — auto-accreditation is architecturally excluded (§3, authority chain).¹⁵

The same generate-verify-re-authorize sequence applies, *mutatis mutandis*, to the other trigger events enumerated in §3: persona update, envelope expansion, and substrate reconfiguration. The fine-tune walkthrough is canonical; the remaining triggers vary in attestation schema but not in process shape.

Why: A paper titled “Continuous-Validity Authorization” must show the authorization process running, not merely name its phases. The walkthrough makes the generate-verify-re-authorize loop testable against a concrete scenario. This is an architectural-specification paper; its sections must specify, not gesture.

§X.3. Defect: §5 Skipjack coupling is label-affixed, not load-bearing

v1.0 said: “**Context integrity** is enforced through accreditation baseline verification. **Structured routing** uses accreditation status as a routing input. **HITL control points** fire on re-authorization triggers. **Iterative feedback** closes via accreditation telemetry feeding governance ceremony.” (§5, complete substantive text.)

The problem: The Round-2 stamp finding (§5 substance audit) identified that each Skipjack mechanism gets one sentence mapping to a continuous-validity concept, but no sentence explains *what changes* about the mechanism when the accreditation surface is continuous rather than snapshot. The section declares a coupling without specifying the delta. Without the delta, §5 is a label, not a specification.

Canonical correction: §5 is expanded to show, for each of the four Skipjack mechanisms, the specific behavioral difference between snapshot-

¹⁴Nicolas Chaillan and Mark Tatum, “Continuous Authority to Operate (cATO): The DoD’s Path to Software Agility,” *IEEE Software* 39, no. 1 (2022): 12–18, <https://doi.org/10.1109/MS.2021.3127155>.

¹⁵National Institute of Standards and Technology, *Risk Management Framework for Information Systems and Organizations*, NIST SP 800-37 Rev. 2 (Gaithersburg, MD, 2018), <https://doi.org/10.6028/NIST.SP.800-37r2>.